Activity 7 - Construct the Lumina 400 Lifecycle Inventory

Alignment: K3 | K4 | K5 | A1 | A3 | LO3 Duration: 50 minutes Objective: Build a complete, metadata-rich lifecycle inventory from the approved product boundary and configuration.

Scenario

The Lumina 400 uses an aluminium housing, glass diffuser, circuit board, LED chips, driver, wiring and packaging. Supplier data ranges from verified product footprints to generic spend estimates. Tuas assembly records electricity and scrap, regional distribution uses road and sea freight, use-phase electricity varies by market, and end-of-life recovery rates are uncertain.

Materials

- Product configuration and bill of materials
- Supplier, production, transport and use-phase records
- Background-dataset catalogue
- LCI and data-quality workbook

Questions

- Which unit processes and flows are needed?
- What is foreground versus background data?
- How will metadata and data quality be captured?
- Which balances and reconciliations can reveal gaps?

Detailed procedure

1. Confirm the functional unit, assessed configuration, lifecycle boundary and allocation rules from Activity 4.
2. Break each lifecycle stage into decision-relevant unit processes and stable process IDs.
3. List material, energy, water, transport, packaging, co-product, waste and emission flows for every process.
4. Capture quantity, unit, source, owner, period, geography, technology, collection method and approval metadata.
5. Classify each value as measured, supplier-specific, industry average, proxy, spend-based or assumption.
6. Map foreground values to controlled background factors or datasets without hiding the link.
7. Perform mass, energy, production and scrap reconciliations where data permits.
8. Rate reliability, completeness, temporal, geographical and technological representativeness.
9. Create a gap-improvement plan for high-impact weak data before final reporting.

Required evidence

- LCI data dictionary
- Unit-process and flow register
- Completed lifecycle inventory

- Data-quality and gap-improvement plan

Acceptance check

Every included unit process has traceable inputs and outputs; metadata and quality ratings are complete; totals reconcile where possible; high-impact gaps have owned improvement actions.

Files

- Activity-07-Workbook.xlsx - editable working file
- Activity-07-Activity-Pack.pdf - printable scenario, questions, procedure and checklist

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